STATS 200A Lecture 11

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Definition 11.1 (Conditional Probability)
For a Borel set B,

$$\boxed{\mathbb{P}(X \in B \mid Y) := \mathbb{E}[\mathbf{1}_B(X) \mid Y]}$$

Definition 11.2 (Proability Kernel)
A Kernel from $(E, \mathcal{E})$ to $(F, \mathcal{H})$ is a map $K : E \times \mathcal{H} \to [0, 1]$ such that

1. $K(y, \cdot)$ is a probability measure for each $y \in E$;
2. $y \mapsto K(y, B)$ is $\mathcal{E}$-measurable for each $B \in \mathcal{H}$

Theorem 11.3 (Existence Stated)
For real X, Y, there is a kernel K from $\mathbb{R}$ to $\mathbb{R}$ such that, for every bounded Borel h,

$$\boxed{\mathbb{E}[h(X) \mid Y] = g_h(Y) \text{ almost surely}, \qquad g_h(y) := \int h(x) K(y, dx).}$$

The kernel is unique $\mathbb{P}_y$-almost everywhere as a probability measure.

Remark 11.4 (Candidate Conditional Law)

$$K(y, \cdot) = \frac{1}{2}\delta_{\sqrt{y}} + \frac{1}{2}\delta_{-\sqrt{y}}, \qquad 0 \le y \le 1$$

Remark 11.5 (The mean alone misses the information)

$$\mathbb{E}[X \mid Y] = 0, \qquad \mathbb{E}[X^2 \mid Y] = Y$$

A constant conditional mean does not imply independence.